



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

CRYPTOGRAPHY AND NETWORK SECURITY [IT-4001]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Given the hex code of the state post initial add round key {EA 83 5C F0 04 45 33 2D 65 5D 98 AD 85 96 B0 C5} answer the following by applying the functions of Advanced Encryption Standard. Refer to the tables 1A (a) and 1A (b). (5)
- A)
1. Show the original State displayed as 4X4 matrix . Show the value of the State after SubBytes.
 2. Show the value of the State after ShiftRows.
 3. Given the output of Mixcolumns post ShiftRows is {47 37 94 ED 40 D4 E4 A5 A3 70 3A A6 46 9F 42 BC} and key constant value {AC 77 63 F3 19 FA DC 21 28 D1 29 41 57 5C 00 4A} compute the add round key.
 4. For the key stream given $W_8 - W_{11}$ {D2 C9 6B B7 49 80 B4 5E DE 7E C6 61 E6 FF D3 C6}, using Key Expansion method compute the complete round key $W_{12} - W_{15}$

Table 1A(a) : RCON Constants

Round	Constant (RCon)	Round	Constant (RCon)
1	(01 00 00 00) ₁₆	6	(20 00 00 00) ₁₆
2	(02 00 00 00) ₁₆	7	(40 00 00 00) ₁₆
3	(04 00 00 00) ₁₆	8	(80 00 00 00) ₁₆
4	(08 00 00 00) ₁₆	9	(1B 00 00 00) ₁₆
5	(10 00 00 00) ₁₆	10	(36 00 00 00) ₁₆

Table 1A(b) : Sub Bytes

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

B) Using the following Symmetric key ciphers, perform the encryption for the text "MIT Bangalore". Use 'X' for padding.

- Affine cipher with the key (7, 2). (5)

Rail fence cipher with key =4.

- 2) Answer the following with respect to RSA cryptosystems: In a public-key system using RSA, you intercept the cipher text $C = 10$ sent to a user whose public key is $e = 5$, $n = 35$. What is the plaintext M ? How to understand the CIA triad in financial systems, appropriate cryptographic methods for PIN confidentiality, and strategies to balance availability across components with different levels of criticality. (5)
- A)
- B) Alice and Bob were two cryptographers tasked with establishing a secure communication channel across an unsafe network using the Diffie-Hellman key exchange protocol. They employ the prime number p : 11 and the Generator number α : 7. The private keys in question are Bob's private key b : 6 and Alice's private key a : 3. Calculate their common secret key. Using the shared secret key decrypt the message "jvwmslaunugplktypyxk" using multiplicative cipher. (5)
- 3) List the main features of the SHA-512 cryptographic hash function and brief the kind of compression function is used in SHA-512? (5)

- A) What is the number of padding bits if the length of the original message is 2590 bits in SHA512 ?
- B) Define Kerberos and name its servers. Briefly explain the duties of each server with neat diagram. (5)
- 4) In a Kerberos-secured environment, Alice, a user, requests access to resources on Bob's server. This process involves interactions with an Authentication Server (AS), a Ticket-Granting Server (TGS), and Bob's Real Server, completed in six steps.
- A) Using a flow diagram, illustrate the sequence and interdependence of each server in the process. Then, evaluate how each server's role contributes to the security of Alice's access request and why this layered authentication is critical. (5)
- B) Consider a financial institution that handles large amounts of sensitive customer information and faces daily cyber threats. In this situation, evaluate the important role that firewalls play in maintaining network security. What are the main functions of firewalls, and how do they help protect the organization's network? (5)
- 5) You are a cybersecurity consultant tasked with designing a comprehensive **copy protection (CP) system** for a new digital home network called "**HomeSecure.**"
- A) What Digital Rights Management technical solutions would you use to provide copy protection? Elaborate this with CSSV and WAP. (5)
- B) Compare and contrast the three categories of intruders in terms of their access levels and the nature of their activities within a computer system. How does each type of user pose different risks to network security? How Okta is used in LinkedIn SSO. (5)

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